cracked main and side rods not permitted. Bearings and bushings shall so fit the rods as to be in a safe and suitable condition for service, and means be provided to prevent bushings turning in rod. Straps shall fit and be securely bolted to rods. The total amount of side motion of rods on crank pins shall not exceed ¼ in. On locomotives used in road service the bore of the main rod bearings shall not exceed pin diameters more than 3/32 in. at front end, or 3/32 in. for back end. The total lost motion at both ends shall not exceed 5/32 in. The bore of side rod bearing on main pin shall not exceed pin diameter more than 3/16 in. * * * *

LIGHTS

Rule 29.—Each locomotive used in road service between sunset and sunrise shall have a headlight which will enable persons with normal vision in the cab of the locomotive, under normal weather conditions, to see a dark object the size of a man for a distance of 1,000 *feet or more ahead of the locomotive; and such headlights must be maintained in good condition. Locomotives used in road service, which are regularly required to run backward for any portion of their trip, except to pick up a detached portion of their train, or in making terminal improvements, shall have on the rear a head-light which will meet the foregoing requirements. Nothing in the foregoing rules shall prevent the use of a device whereby the light may be diminished in yards and at stations to an extent that will enable the person or persons operating the locomotive to see a dark object the size of a man for* 300 *feet or more ahead of the locomotive under the same conditions as set forth above. When two or more locomotives are used in the same train, the leading locomotive only, will be required to display a head-light. (Rule 29 was objected to by the committee representing the carriers.)*

Rule 31.—Each locomotive used in yard service between sunset and sunrise shall have two headlights, one located on the front of the locomotive and one on the rear, each of which will enable persons with normal vision, in the cab of the locomotive, under normal weather conditions, to see a dark object the size of a man for a distance of 300 *ft. or more; and such headlights must be maintained in good condition. (Rule* 31 *was objected to by the committee representing the carriers.)*

RUNNING GEAR

Rule 33.—Driving, trailing and engine truck axles with any of the following defects shall not be continued in service: Bent axles; cut journals that cannot be made to run cool without turning; seamy journals in steel axles; transverse seams in iron axles, or any seams in iron axles causing axles to run hot; or unsafe on account of usage, accident or derailment; main driving, trailing truck or engine truck axles more than one-half inch under original diameter; other driving axle more than three-fourth inch under original diameter.

Rule 34.—Diameter of tender truck axle journals shall be governed by axle loads. The minimum diameter of journals for various axle loads shall be as follows:

Axle load.	Minimum diameter of journals.
50,000 lb.	5½ in.
38,000 lb.	5 in.
31,000 lb.	4½ in.
22,000 lb.	3¾ in.
15,000 lb.	3¼ in.

Rule 35.—Tender truck axles with any of the following defects shall not be continued in service: Bent axle; cut journals that cannot be made to run cool without turning; seamy journals in steel axles or transverse seams in journals of iron axles, or unsafe on account of usage, accident or derailment; collars broken or worn to one-fourth inch or less in thickness; fillet in back shoulder worn out.

Rule 36.—Crank pins shall be securely applied. Shimming or prick punching crank pins will not be allowed. All crank pins allowed after January 1, 1916, shall have the date applied and kind of material used legibly stamped on end of pin. Crank pin collars and collar bolts shall be maintained in a safe and suitable condition for service.

Rule 37.—Driving boxes shall be maintained in a safe and suitable condition for service. Broken and loose bearings shall be renewed; not more than one shim may be used between box and bearing.

Rule 40.—The total lateral motion or play between the hubs of the wheels and the boxes on any pair of wheels shall not exceed the following limits:

For engine truck wheels (trucks with swing centers)	1 in.
For engine truck wheels (trucks with rigid centers)	1½ in.
For tra ling truck wheels	1 in.
For driving wheels (more than one pair)	¾ in

These limits may be increased on locomotive operating on track where the curvature exceeds 20 deg. when it can be shown that conditions require additional lateral motion. The lateral motion shall in all cases be kept within such limits that the driving wheels, rods or crank pins will not interfere with other parts of the locomotive.

Rule 42.—Springs and equalizers shall be arranged to insure the proper distribution of weight to the various wheels of the locomotive; maintained approximately level, and in a safe and suitable condition for service. Springs or spring rigging with any of the following defects shall be renewed or properly repaired: One long leaf or two or more shorter leaves broken; springs with leaves working in band; broken coil springs; broken driving box saddle, equalizer, hanger, bolt or pin.

Rule 43.—The minimum height of flange for driving and trailing wheel tires for locomotives used in road service, measured from tread, shall be 1 in. except for locomotives originally constructed for plain tires when the minimum height of flange on one pair of wheels may be ⅞ in. The maximum taper for tread of tires from throat of flange to outside of tire for driving and trailing wheels for locomotives used in road service shall be ¼ in. and for locomotives used in switching service 5/16 in. The minimum width of tires for driving and trailing wheels of standard gage locomotives shall be 5½ in. for flanged tires and 6 in .for plain tires . . . . When all tires are turned or new tires applied to driving and trailing wheels of locomotives, the diameter of the wheels on the same axle or in the same driving-wheel base shall not vary more than 3/32 in. When a single tire is applied the diameter must not vary more than 3/32 in. from that of the opposite wheel on the same axle. When a single pair of tires is applied the diameter must be within 3/32 in. of the average diameter of the wheels in the driving-wheel base to which they are applied. Driving and trailing wheel tires with any of the following defects shall not be continued in service: Slid flat spot 2½ in. or more in length; flange 15/16 in. or less in thickness, gaged at a point ⅜ in. above the tread, or having flat vertical surface 1 in. or more from tread; tread worn hollow 5/16 in. on locomotives used in road service or ⅜ in. on locomotives used in switching service; flange more than 1½ in. from tread to top of flange.

Rule 46.—Wheels shall be securely pressed on axles. Prick punching or shimming the wheel fit will not be permitted. The diameter of wheels on the same axle shall not vary more than 3/32 in. Wheels used on standard gage track will be out of gage if the inside gage of flanges, measured on base line, is less than 53 in. or more than 53⅜ in. The distance back to back of flanges of wheels mounted on the same axle shall not vary more than ¼ in.

Rule 47.—Cast-iron or cast-steel wheels with any of the following defects shall not be continued in service:

Slid Flat.—When the flat sport is 2½ in. or over in length, or if there are two or more adjoining spots each 2 in. or over in length.

Broken or Chipped Flange.—If the chip exceeds 1½ in. in length and ½ in. in width.

Broken Rim.—If the tread, measured from the flange at a point ⅝ in. above the tread, is less than 3¾ in. in width.

Shelled Out.—Wheels with defective treads on account of cracks or shelled out spots 2½ in. or over, or so numerous as to endanger the safety of the wheel.

Brake Burn.—Wheels having defective treads on account of cracks or shelling out due to heating.

portance as an efficient distribution of steam in the cylinders. If the valves are not properly balanced an enormous stress is imposed upon the valve gear and transmitting rods in the work they are called upon to do. In calculating the dimensions of stems and rods the designer is obliged to consider the work that would have to be performed in case of an accident to the balancing strips and lubricating apparatus.

Piston Valves.—The advantages of the piston valve are as follows:

Increased port area for both admission and exhaust.

Ports in cylinder made very straight and direct.

A simpler, lighter, and cheaper cylinder casting; and a wearing face separate from the cylinder casting that can be cheaply renewed.

Its adaptability to any design of valve gear, since it can be placed above the cylinder, between the frame rails, or in any other position with equal facility.

With inside admission valves the steam passages are better protected from the cold and radiation, and the steam chest heads and packings relieved from all pressure except that of the exhaust steam, which is but a few pounds above that of the atmosphere and so puts, really, very little stress upon these parts.

Accessibility of parts. Peep hole plugs make it possible to get port marks without removing valve chamber heads.

Better balance which makes it easier to handle, and decreases the wear and tear on the motion work.

The relative frictional resistance of the piston valve is much less than the slide valve. (This was proven by tests made on the C. B. & Q.).

The greatest disadvantage under which the piston valve labors is its inability to relieve excess pressure in the cylinder port by lifting, after the manner of the slide valve. Many of the weak features of the piston valve have been eliminated in the recent designs, and where proper care is taken in the maintenance and operation of piston valves they give splendid service.

Piston valve cylinders have considerable advantage over slide valve cylinders. The walls lend themselves more readily to curved lines than do those of the slide valve. (The avoidance of flat walls exposed to steam pressure is one of the most important things to be considered in cylinder work). As piston valves can be made of any desired length the steam ports can be made very short and direct. For this reason also the steam chest should be placed as close to the bore as will allow the barrel flange to be turned for the head casing. The piston valve cylinder and all its appurtenances weigh somewhat less than a first class slide valve and its appurtenances of equal capacities. Piston valves give flexibility to the design because they can be located in almost any position with regard to the cylinders.

As regards the size of piston valve that should be used for large power there is considerable difference of opinion. Exhaustive tests made on the Pennsylvania Railroad showed that the piston valve could be largely standardized, and that a 12-in. diameter of valve was large enough for cylinders up to 27 in. in diameter, when used with superheated steam. The standard 12-in. diameter of valve was then developed, and with only one change in the overall length, it now fits locomotives of 14 different classes on that road.

Valve Gears.—At the present time four general designs of valve gears are used in this country. They are: The Stephenson link motion; the Walschaert valve gear; the Baker valve gear, and the Southern valve gear.

For many years the Stephenson link motion was used almost exclusively in this country, but with the introduction of the very heavy power, for structural reasons, some form of outside valve gear became imperative. The principal advantages of outside valve gears are: Accessibility for lubrication, inspection and repairs; opportunity for heavy cross bracing between the frames; motion can be made more direct, and with fewer wearing parts, and ability to hold their adjustment for a longer length of time.

The disadvantages of the outside valve gears are: Liability of damage from side swipes; rod work, changing of tires, etc. made more difficult due to the location of outside valve gears, and slight distortion of the valve events due to up and down movement of the main wheel, and with it the eccentric crank, and back end of the eccentric rod.

Aside from the matter of good steam distribution, there are a number of items that must be considered in the selection of a valve gear for different classes of service. They are as follows: First cost; cost of maintenance and repairs; efficiency and reliability of service; ability to hold adjustment, and ease of handling in the cab.

Three of the most important features to be considered, in the design and construction of an outside valve gear are: Necessity for providing as rigid a support as possible; in the provision for reducing wear to a minimum, and in providing sufficient lubrication at every point.

The advantages to be derived from the high class materials for valve gear parts have been recognized, and at the present time the tendency is to use steel for these parts. The recent Pennsylvania engines have all motion work parts of heat-treated steel. Soft steel is used extensively for transmitting and other valve gear rods, and the jaws are usually case-hardened to give good wearing qualities. Cast steel is only adapted for certain parts, and where used should be annealed. The American Locomotive Company follows the practice of using drop forged motion work parts wherever it is possible to do so. Drop forged motion work parts have the advantage of requiring very little finish, and are strong and durable.

The outside valve gears produce a more uniform steam distribution with a lower percentage of preadmission than the Stephenson link motion. They hold their adjustment, and consequently give a better steam distribution for a longer length of time. Constant lead is the characteristic of all well known outside valve gears. While there is no reliable data available as to the cost of maintenance of the outside valve gears, in comparison with the Stephenson link motion, it is generally considered that the cost of maintenance of the outside gears is from 60 to 75 per cent less than for the Stephenson gear.

The Stephenson gear possesses the peculiarity of being exceedingly sensitive to a close adjustment of all its parts in order that a correct action and proper distribution of the steam may be obtained. It is the most flexible of any in use and can be most readily adapted to irregularities in the running and operation of the engine. At the same time it will get out of adjustment very easily, and requires the utmost care in its design in order that it may work properly. With the link motion in actual service there are three sources of error which cause a variation of the same events for the two ends of the cylinder, and which must be compromised for in some manner. They are: The location of the eccentric rod pins back of the link arc; the angular vibration of the eccentric rods, and the angular vibration of the connecting rod. To a certain extent the latter two compensate the first, but not entirely, and to complete the compensation the hanger stud is set back of the link arc.

In certain class of service the variable lead given by the Stephenson link motion permits the locomotive to accelerate more rapidly and to better adjust itself to different operating conditions than is possible with the valve gears giving a constant lead. Variable lead and flexibility are the characteristics of the gear. An added advantage of the Stephenson motion is that everything is inside where it is well protected from damage by a side swipe. For this reason several roads are still specifying the Stephenson gear for moderate sized switch engines. Switch engines are especially apt to be side swiped in congested yards.

On the other hand, because of the weight and power of modern locomotives, it is almost impossible to get a satisfactory design of the Stephenson link motion between the frames.

[This report also included a very clear description of the